

Simulating the Real World: A Unified Survey of Multimodal Generative Models

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Abstract—Understanding and replicating the real world is a critical challenge in Artificial General Intelligence (AGI) research. To achieve this, many existing approaches, such as world models, aim to capture the fundamental principles governing the physical world, enabling more accurate simulations and meaningful interactions. However, current methods often treat different modalities, including 2D (images), videos, 3D, and 4D representations, as independent domains, overlooking their interdependencies. Additionally, these methods typically focus on isolated dimensions of reality without systematically integrating their connections. In this survey, we present a unified survey for multimodal generative models that investigate the progression of data dimensionality in real-world simulation. Specifically, this survey starts from 2D generation (**appearance**), then moves to video (**appearance+dynamics**) and 3D generation (**appearance+geometry**), and finally culminates in 4D generation that integrate all dimensions. To the best of our knowledge, this is the first attempt to systematically unify the study of 2D, video, 3D, and 4D generation within a single framework. To guide future research, we provide a comprehensive review of datasets, evaluation metrics, and future directions to foster insights for newcomers. This survey serves as a bridge to advance the study of multimodal generative models and real-world simulation within a unified framework.

Index Terms—Generative models, image generation, video generation, 3D generation, 4D generation, deep learning, literature survey.

I. INTRODUCTION

FOR decades, the research community has aspired to develop systems that encapsulate the fundamental principles of the physical world, a cornerstone in the journey towards Artificial General Intelligence (AGI) [1]. Central to this endeavor is simulating the real world with machines,

aiming to capture the complexities of reality through the lens of multimodal generative models. The resultant world simulator holds the promise of advancing the understanding of the real world, unlocking transformative applications such as virtual reality [2], games [3], robotics [4], and autonomous driving [5].

The term “world simulator” was first introduced by Ha David [6], drawing an analogy to the concept of a mental model [7] in cognitive science. Building on this perspective, modern researchers [8] formulate the simulator as an abstract framework that enables intelligent systems to simulate the real world through multimodal generative models. These models encode visual contents and spatial-temporal dynamics of the real world into compact representations. As geometry, appearance, and dynamics jointly contribute to the realness of generated contents, these three aspects are widely investigated by the community [9]. Traditional real-world simulation methods have long relied on graphics techniques that incorporate geometry, texture, and dynamics. Specifically, geometry and texture modeling [10] are employed to create the objects, while methods like keyframe animation [11] and physics-based simulation [12] are adopted to simulate the movement and behavior of objects over time.

Despite great progress, these traditional methods often require extensive manual designs, heuristic rule definitions, and computationally expensive processing, limiting their scalability and adaptability to diverse scenarios. Recently, learning-based approaches, particularly multimodal generative models, have revolutionized content creation by providing a data-driven approach to realistic simulations. These approaches reduce the reliance on manual efforts, improve generalization across tasks, and enable intuitive interactions between humans and models. For example, Sora [13] has garnered significant attention for its realistic simulation capabilities, demonstrating an early-stage understanding of physical laws. The emergence of such generative models introduces new perspectives and methodologies, addressing the limitations of traditional methods by reducing the need for extensive manual design and computationally expensive modeling while enhancing adaptability and scalability in diverse simulation scenarios.

Though existing generative models offer powerful techniques for synthesizing realistic content in distinct data dimensions, the real world exhibits inherently high-dimensional complexity, and a comprehensive review that systematically integrates these advancements across different dimensions is

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A project associated with this survey is available at <https://github.com/ALEEEHU/World-Simulator>.

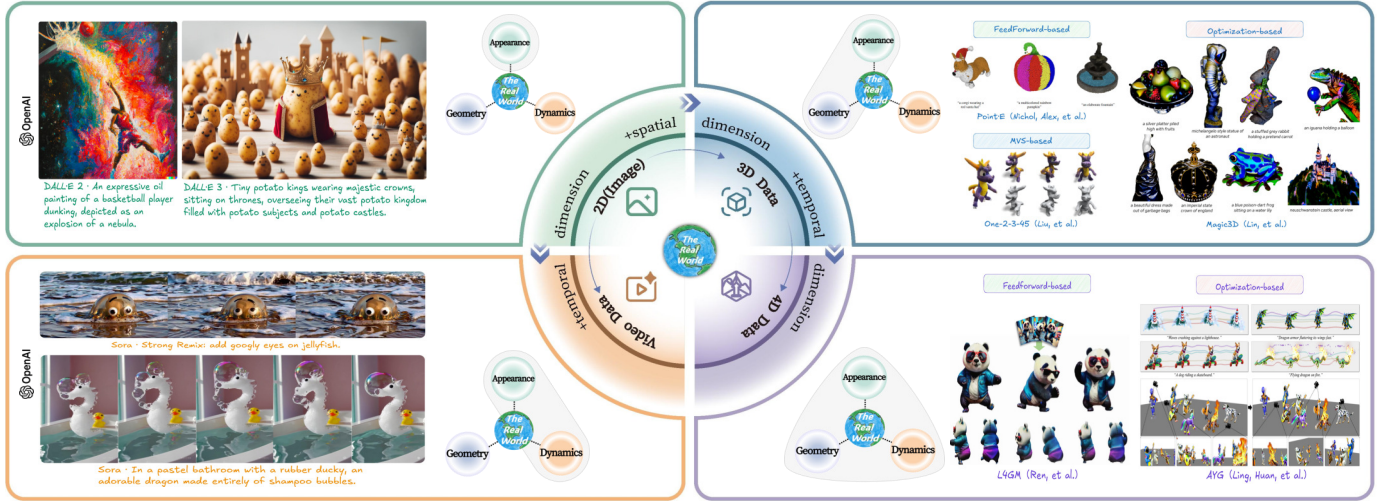


Fig. 1: Roadmap of dimensional growth from 2D images to video, 3D, and 4D content in real-world simulation, outlining a conceptual taxonomy based on the coverage of data properties (i.e., appearance, geometry, and dynamics).

still absent. This survey aims to bridge this gap by unifying the study of real-world simulation from the perspective of data dimension growth, as illustrated in Figure 1. Specifically, we start from 2D generation (appearance only) and then extend it to video and 3D generation by incorporating dynamics and geometry dimensions, respectively. Finally, we culminate in 4D generation by integrating all dimensions.

In summary, this survey makes three key contributions. *First*, it provides a systematic review of methods for real-world simulation from the perspective of data dimension growth through the lens of multimodal generative models. To the best of our knowledge, this is the first survey that unifies the study of 2D, video, 3D, and 4D generation, offering a structured and comprehensive overview of this research area. *Second*, it surveys the commonly used datasets, their properties, and the corresponding evaluation metrics from various perspectives. *Third*, it identifies open research challenges, aiming to guide further exploration in this field.

Previous surveys on generation models mainly concentrates on text-to-image, text-to-video, and text-to-3D generation separately without comprehensively studying their relationships. In contrast, this survey seeks to provide a more integrated perspective on multimodal generative models by tracing how generative models have evolved from handling appearance alone (2D generation), to incorporating dynamics (video generation) and geometry (3D generation), and ultimately to integrating appearance, dynamics, and geometry in 4D generation. This dimensional perspective aims to bridge previously isolated research areas and highlight common challenges and opportunities across them.

We envision this survey as a resource that provides valuable insights for newcomers and fosters critical analysis among experienced researchers. The survey is organized as follows. Section II introduces the foundational concepts of deep generative models. Section III presents four key paradigms: 2D, video, 3D, and 4D generation. Section IV reviews datasets and evaluation metrics for these paradigms. Finally, Section V outlines future directions, and Section VI concludes the survey.

II. PRELIMINARIES

Deep generative models learn complicated and high-dimensional data distributions with the aid of deep neural networks. Denote the data sample as $\mathbf{x}$ and its distribution as $p_{data}(\mathbf{x})$, the objective of deep generative models is to approximate the $p_{data}(\mathbf{x})$ with $p_{\theta}(\mathbf{x})$, where θ is the parameter of models. In this section, we briefly review several mainstream generative models (Table I), including generative adversarial networks (GANs) [14], variational autoencoders (VAEs) [15], autoregressive models (AR Models) [16], normalizing flows (NFs) [17], and diffusion models [18].

A. Generative Adversarial Networks (GANs)

GANs avoid the parametric form of $p_{\theta}(\mathbf{x})$ but represent $p_{\theta}(\mathbf{x})$ as the distribution of samples produced by a generator. It has been shown that $p_{\theta}(\mathbf{x})$ will converge to $p_{data}(\mathbf{x})$ under certain conditions [14].

Specifically, the generator takes a noise vector $\mathbf{z}$ as input to synthesize a data sample $G(\mathbf{z}; \theta_g)$. The $p_{\theta}(\mathbf{x})$ is defined as the distribution of $G(\mathbf{z}; \theta_g)$, where $\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})$. Meanwhile, the discriminator $D(\mathbf{x}; \theta_d)$ identifies whether an input data sample is a real one or a synthetic one. During training, the discriminator is trained to distinguish the generated samples from real data, while the generator is trained to deceive the discriminator. The process can be formulated as:

$$\min_G \max_D V(G, D) = \mathbb{E}_{\mathbf{x} \sim p_{data}(\mathbf{x})} [\log D(\mathbf{x}; \theta_d)] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log (1 - D(G(\mathbf{z}; \theta_g); \theta_d))]. \quad (1)$$

There are several challenges associated with GAN training. For example, the Nash equilibrium may not always exist [19] or be hard to achieve [20], resulting in unstable training. Another problem is mode collapse, where the generator produces only specific types of samples with low diversity [20], [21].

TABLE I: Comparison of deep generative models.

Model Types	Advantages	Disadvantages
GANs [14]	1) Flexible 2) Efficient inference	1) Unstable training 2) Lack diversity 3) Mode collapse
VAEs [15]	1) Data compression	1) Posterior collapse 2) Blur
AR Models [16]	1) Explicit density	1) Markov assumption 2) Difficult to parallelize
NFs [17]	1) Explicit density	1) Limited capacity 2) Restricted architecture
Diffusion Models [18]	1) High-quality samples 2) Learn complex distribution	1) Expensive computation

B. Variational Autoencoders (VAEs)

The variational autoencoders (VAEs) formulate the $p_\theta(\mathbf{x})$ as,

$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x}|\mathbf{z})p_\theta(\mathbf{z})d\mathbf{z}, \quad (2)$$

where $p_\theta(\mathbf{z})$ is a prior distribution of $\mathbf{z}$, and $p_\theta(\mathbf{x}|\mathbf{z})$ is the distribution of $\mathbf{x}$ conditioned on $\mathbf{z}$. However, since this integration is usually intractable, VAEs maximize the lower bound of $\log p_\theta(\mathbf{x}) \geq -\text{KL}(q_\theta(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z})) + \mathbb{E}_{q_\theta(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]$, where $\text{KL}(q_\theta(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}))$ is the KL divergence between $q_\theta(\mathbf{z}|\mathbf{x})$ and $p_\theta(\mathbf{z})$, and $\mathbb{E}_{q_\theta(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]$ is computed by the Stochastic Gradient Variational Bayes estimator [15]. The challenges for VAEs include undesirable stable equilibrium, Blurriness and so on [22].

C. Autoregressive Models (AR Models)

Autoregressive (AR) models factorize $p_\theta(\mathbf{x})$ as a product of conditional probabilities,

$$p_\theta(\mathbf{x}) = p(x_1, \dots, x_d) = \prod_{i=1}^d p_\theta(x_i | x_1, \dots, x_{i-1}), \quad (3)$$

where d is the sequence length. Such factorisation simplifies multivariate density estimation and has been widely adopted to model pixels sequentially in images [23]–[25].

To reduce the quadratic cost of attention in standard transformer-based AR models, several non-transformer architectures have recently been introduced. RWKV [26], Mamba [27], and RetNet [28] replace or augment attention with recurrent or state-space mechanisms. RWKV and Mamba employ purely recurrent designs that maintain a fixed-size memory, delivering linear-time inference over moderate sequence lengths but still facing challenges at extreme context sizes. RetNet updates hidden states through a retention mechanism, providing an efficient alternative to global self-attention. Although these architectures show promising results on language and other sequence tasks, their use as backbones for deep generative models is still limited. Future work that integrates them into generative pipelines may improve the trade-off among sample quality, scalability, and memory usage.

D. Normalizing Flows (NFs)

NFs employ an invertible neural network $g(\cdot)$ to map $\mathbf{z}$ from a known and tractable distribution to the real data distribution. In this way, $p_\theta(\mathbf{x})$ can be formulated as,

$$p_\theta(\mathbf{x}) = p(f(\mathbf{x}))|\det J(g(g^{-1}(\mathbf{x})))|^{-1}, \quad (4)$$

where $g^{-1}(\cdot)$ is the inverse of $g(\cdot)$, and $J(g(\mathbf{z})) = \frac{\partial g}{\partial \mathbf{z}}$ is the Jacobian of $\mathbf{g}$. NFs construct more complicated non-linear invertible functions by compositing a set of N bijective functions and define $g = g_N \circ \dots \circ g_1$.

E. Diffusion Models

Diffusion models are a class of probabilistic generative models that iteratively corrupt data by introducing noise and subsequently learn to reverse this process to generate samples. We define $p_\theta(\mathbf{x})$ with an energy term $s_\theta(\mathbf{x})$,

$$p_\theta(\mathbf{x}) = \exp(-s_\theta(\mathbf{x}))/Z_\theta, \quad (5)$$

where $Z_\theta = \int \exp(-s_\theta(\mathbf{x}))d\mathbf{x}$ is the normalization term. Since Z_θ is intractable to compute, the diffusion models learn the score function $\nabla_{\mathbf{x}} \log p_\theta(\mathbf{x}) = -\nabla_{\mathbf{x}} s_\theta(\mathbf{x})$ instead.

The forward process that transforms data distribution to standard Gaussian is defined by $d\mathbf{x} = f(\mathbf{x}, t)dt + g(t)d\mathbf{w}$, where $f(\mathbf{x}, t)$ is the drift coefficient, $g(t)$ is the diffusion coefficient, $\mathbf{w}$ is the standard Wiener process, and $t \in [0, 1]$. The samples are generated by the corresponding reverse process, which is described by $d\mathbf{x} = [f(\mathbf{x}, t) - g^2(t)\nabla \log p_t(\mathbf{x})]dt + g(t)d\bar{\mathbf{w}}$, where $\bar{\mathbf{w}}$ is a standard Wiener process when time flows backward from 1 to 0.

III. PARADIGMS

This section presents methods for simulating the real world from the perspective of data dimension growth. It begins with 2D generation (Section III-A) for appearance modeling and then move to video generation (Section III-B) and 3D generation (Section III-C) by incorporating dynamics and geometry dimensions. Finally, by integrating all these three dimensions, recent advances in 4D generation (Section III-D) are presented.

A. 2D Generation

Recently, significant advancements have been made in the field of generative models, particularly in text-to-image generation. Text-to-image generation has attracted attention for its capability to produce realistic images from textual descriptions by capturing the appearance of the real world. Utilizing techniques like diffusion models, large language models (LLMs), and autoencoders, these models achieve high-quality and semantically accurate image generation.

1) *Algorithms*: **Imagen** [29] builds on the principles established by GLIDE but introduces significant optimizations and improvements. Instead of training a task-specific text encoder from scratch, Imagen uses pre-trained and frozen language models and reduces computational demands. Imagen tested models trained on image-text datasets (e.g., CLIP [30]) and models trained on pure text datasets (e.g., BERT [31] and T5 [32]). This practice shows that scaling up language models enhances image fidelity and text congruence more effectively than enlarging image diffusion models.

DALL-E [33] (version 1) uses a transformer architecture that processes both text and images as a single stream of data. DALL-E 2 [34] utilizes the powerful semantic and stylistic capabilities of CLIP [30], which employs a generative diffusion decoder to reverse the process of the CLIP image encoder. DALL-E 3 [35] builds upon the advancements of DALL-E 2 [34], offering significant improvements in image fidelity and text alignment. It enhances text understanding, allowing for more accurate and nuanced image generation from complex descriptions. DALL-E 3 is integrated with ChatGPT [36], enabling users to brainstorm and refine prompts directly within the ChatGPT interface, which simplifies the process of generating detailed and tailored prompts. The model produces images with higher realism and better alignment to the provided text, making it a powerful tool for both creative and professional applications.

DeepFloyd IF [37] is celebrated for its exceptional photorealism and advanced language understanding. This system is modular, featuring a static text encoder and three sequential pixel diffusion modules. Initially, the base model creates 64×64 pixel images from textual descriptions. These images are then enhanced to 256×256 pixels and further to 1024×1024 pixels by two super-resolution models. Each phase utilizes a static text encoder derived from the T5 [32] transformer to generate text embeddings, which are subsequently processed by a U-Net architecture with integrated cross-attention and attention pooling mechanisms.

Stable Diffusion (SD) [38], also known as Latent Diffusion Model (LDM), enhances training and inference efficiency on limited computational resources while producing high-quality and diverse images. The denoising process takes place in the latent space of pre-trained autoencoders, which map images into a spatial latent space. The underlying U-Net architecture is augmented with a cross-attention mechanism to model the conditional distribution, which can include text prompts, segmentation masks, and more. It used CLIP [30] text embeddings as condition and trained on the LAION [39] dataset to generate images at a resolution of 512×512 (with a latent resolution of 64×64). Building on Stable Diffusion, SDXL [40] employs a U-Net backbone that is three times larger. It introduces additional attention blocks and a larger cross-attention context by utilizing a second text encoder. Additionally, SDXL includes a refinement model that enhances the visual fidelity of samples generated by SDXL through a post-hoc image-to-image technique.

FLUX.1 [41] utilizes a hybrid architecture that integrates multimodal and parallel diffusion transformer blocks, achieving a remarkable scale of 12 billion parameters. By employing flow matching, a straightforward yet effective technique for training generative models, FLUX.1 outperforms prior state-of-the-art diffusion models. The suite also features rotary positional embeddings and parallel attention layers, greatly improving model performance and efficiency.

B. Video Generation

Due to the structural similarities between images and videos, early approaches to video generation predominantly

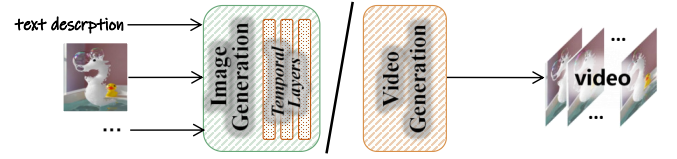


Fig. 2: An illustration of video generation paradigm. Video generation models are constructed on top of image generation models by adding temporal layers or from scratch.

adapted and fine-tuned existing 2D image generation models (Sec. III-A). The initial perspective on this challenge focused on incorporating mechanisms for temporal dynamic modeling, often by adding temporal layers (Fig. 2), such as attention and convolution, to the architecture. A common training strategy also involved mixing 2D images and video data to enhance the visual quality of the generated videos. Inspired by Sora [13], state-of-the-art models now frequently employ a diffusion transformer architecture. These models operate on a compressed spatio-temporal latent space by breaking down video into a series of “patches” that serve as tokens for the transformer. This approach processes both spatial and temporal information simultaneously. Despite their sophistication, these models typically retain the ability to process single images as one-frame videos, allowing them to leverage the vast data available for 2D image generation.

In this section, we classify these models into three main categories based on their underlying generative machine learning architectures. Figure 3 summarizes recent text-to-video generation techniques. For readers seeking a more in-depth exploration, detailed surveys are available [42], [43].

1) Algorithms: (1) VAE- and GAN-based Approaches. Before diffusion models, video generation research has advanced through two primary approaches: VAE-based and GAN-based methods, each contributing unique solutions to the challenges of video synthesis. VAE-based methods evolved from SV2P [44] in stochastic dynamics to a combination of VQ-VAE [71] with transformers in VideoGPT [72], efficiently handling high-resolution videos through hierarchical discrete latent variables. Notable improvements came from the parameter-efficient architecture in FitVid [45] and the integration of adversarial training for more realistic predictions. Parallel developments in GAN-based approaches brought significant innovations, starting with MoCoGAN [46], which decomposes content and motion components for controlled generation. StyleGAN-V [47] advances this by treating videos as time-continuous signals through positional embeddings, while DIGAN [48] introduces implicit neural representations for improved continuous video modeling. StyleInV [49] leverages a temporal style-modulated inversion network with pre-trained StyleGAN [73] generators, marking another milestone in high-quality frame synthesis with temporal coherence.

(2) Diffusion-based Approaches. Text-to-video generation has recently seen significant advancements, with approaches generally falling into two categories: U-Net-based architectures and transformer-based architectures.

(i) U-Net-based Architectures. The pioneering Video Dif-

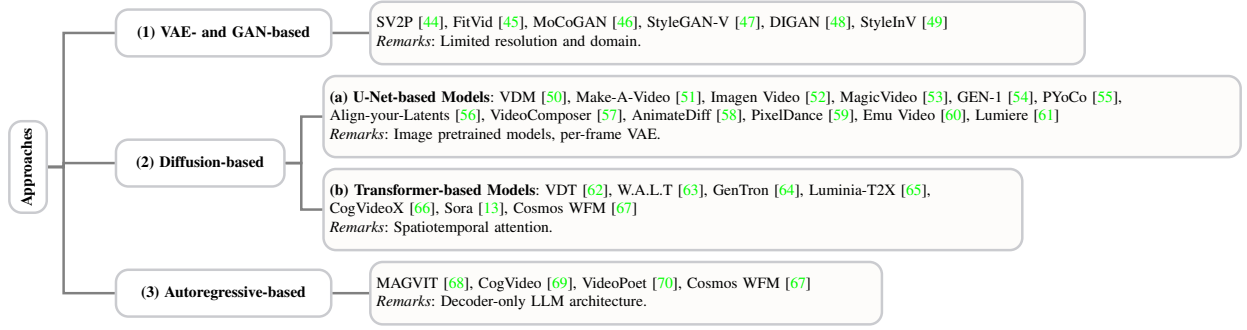


Fig. 3: Overview of text-to-video generation technologies categorized by three main approaches.

fusion Models (VDM) [50] achieved high-fidelity, temporally coherent video generation by extending image diffusion architectures and introducing joint image-video training for reduced gradient variance. Make-A-Video [51] advanced text-to-video generation without paired text-video data by leveraging existing visual representations [30] and innovative spatial-temporal modules. Imagen Video [52] introduced a cascade of diffusion models combining base generation with super-resolution, while MagicVideo [53] achieved efficient generation through latent diffusion in low-dimensional space. GEN-1 [54] focused on structure-preserving editing using depth estimates, while PVoCo [55] demonstrated efficient fine-tuning with limited data through carefully designed video noise priors. Align-your-Latents [56] achieved high-resolution generation (1280×2048) by extending Stable Diffusion [38] with temporal alignment techniques. Show-1 [74] combined pixel-based and latent-based approaches for enhanced quality and reduced computation. VideoComposer [57] introduced a novel paradigm for controllable synthesis through a Spatio-Temporal Condition encoder, enabling flexible composition based on multiple conditions. AnimateDiff [58] presented a plug-and-play motion module with transferable motion priors and introduced MotionLoRA for efficient adaptation. PixelDance [59] enhanced generation by incorporating both first and last frame image instructions alongside text prompts.

(ii) Transformer-based Architectures. Following the success of Diffusion Transformer (DiT) [75], transformer-based models gained prominence. VDT [62] introduced modularized temporal and spatial attention mechanisms for diverse tasks including prediction, interpolation, and completion. W.A.L.T [63] achieved photorealistic generation through a unified latent space and causal encoder architecture, producing high-resolution videos at 512×896 . Snap Video [76] improved training efficiency 3.31 times through spatially and temporally redundant pixel handling, while GenTron [64] scaled to over 3B parameters with motion-free guidance. Luminia-T2X [65] integrated multiple modalities through zero-initialized attention and tokenized latent spatial-temporal space. CogVideoX [66] excelled in long-duration video generation through expert transformers, 3D VAEs, and progressive training, achieving state-of-the-art performance validated by multiple metrics. The groundbreaking Sora [13] is an advanced diffusion transformer model that emphasizes generating high-quality images and videos across different resolutions, aspect

ratios, and durations. Sora achieves flexible and scalable generation capabilities by tokenizing the latent spatial-temporal space.

(3) Autoregressive-based Approaches. Parallel to diffusion-based methods, autoregressive frameworks inspired by large language models (LLMs) have emerged as an alternative approach to video generation. These methods typically follow a two-stage process: first encoding visual content into discrete latent tokens using vector quantized auto-encoders like VQGAN [77] and MAGVIT [68], [78]–[81], then modeling the token distribution in the latent space. CogVideo [69], a 9-billion-parameter transformer model built upon the pre-trained text-to-image model CogView [82], represents a significant advancement in this direction. It employs a multi-frame-rate hierarchical training strategy to enhance text-video alignment and, as one of the first open-source large-scale pre-trained text-to-video models, establishes new benchmarks in both machine and human evaluations. VideoPoet [70] introduces a decoder-only transformer architecture for zero-shot video generation, capable of processing multiple input modalities including images, videos, text, and audio. Following LLM training paradigms with pretraining and task-specific adaptation stages, VideoPoet achieves state-of-the-art performance in zero-shot video creation, particularly excelling in motion fidelity through its diverse generative pretraining objectives.

Evaluation. The evaluation of video generation models has evolved with the task’s increasing complexity. Early approaches relied on distribution-based metrics, most notably the Fréchet Video Distance (FVD) [50], [56], [83]. As a temporal extension of the Fréchet Inception Distance (FID) [84] for images, FVD compares spatiotemporal feature distributions to assess visual quality and coherence. More recent benchmarks, such as VBench [85], offer more granular analysis of specific attributes like motion smoothness and subject identity using features from models like CLIP [30] and DINO [86]. However, because automated metrics often misalign with human perception, the field has increasingly shifted towards human studies for a more holistic and accurate assessment, particularly for advanced open-domain models. Tab. II presents a human preference evaluation of modern video generation models.

2) Applications: **(1) Video editing** has recently benefited significantly from diffusion models, enabling sophisticated modifications while maintaining temporal consistency. The field has evolved through several innovative approaches ad-

Dimension ($\uparrow$)	Hunyuan [87]	Mochi [88]	SoRA [13]	Wan-14B [89]
Motion Quality	0.413	0.420	0.482	0.415
Human Artifacts	0.734	0.622	0.786	0.691
Pixel Stability	0.983	0.981	0.952	0.972
ID Consistency	0.935	0.930	0.925	0.946
Physical	0.898	0.728	0.933	0.939
Smoothness	0.890	0.530	0.930	0.910
Image Quality	0.605	0.530	0.665	0.640
Scene Quality	0.373	0.368	0.388	0.386
Stylization	0.386	0.403	0.606	0.328
Object Accuracy	0.912	0.949	0.932	0.952

TABLE II: Quantitative comparison between different text-to-video methods. Results are obtained from [89].

addressing different aspects of video manipulation. Early developments include Tune-A-Video [90], which pioneered a one-shot tuning paradigm extending text-to-image diffusion models to video generation through spatiotemporal attention mechanisms. Temporal consistency has been addressed through various approaches. VidToMe [91] introduced token merging for aligning frames, while EI [92] developed specialized attention modules. Several works focused on specialized editing capabilities. Ground-A-Video [93] tackled multi-attribute editing through a grounding-guided framework, while Video-P2P [94] introduced cross-attention control for character generation. Recent frameworks like UniEdit [95] and AnyV2V [96] represent the latest evolution, offering tuning-free approaches and simplified editing processes. Specialized applications such as CoDeF [97] and Pix2Video [98] have introduced innovative techniques for temporally consistent processing and progressive change propagation. These methods successfully balance content editing with structural preservation, marking significant advancements in video manipulation technology.

(2) Novel view synthesis has been revolutionized by video diffusion models, which benefit from learned priors on real-world geometry, enabling high-quality view generation from limited input images. ViewCrafter [99] pioneered this direction by integrating video diffusion models with point-based 3D representations, introducing iterative synthesis strategies and camera trajectory planning for high-fidelity results from sparse inputs. Camera control has emerged as a crucial aspect, with CameraCtrl [100] introducing precise camera pose control through a plug-and-play module. Several innovative methods have addressed view consistency challenges. ViVid-1-to-3 [101] reformulated novel view synthesis as video generation of camera movement, while NVS-Solver [102] introduced a zero-shot paradigm that modulates diffusion sampling with given views. This trend shows a convergence toward leveraging video diffusion priors while maintaining geometric consistency and camera control, enabling increasingly realistic synthesis applications.

(3) Human animation in videos has gained significance in video generation, which plays a pivotal role in the world simulator, as discussed in Sec. III-B1. This is particularly important because humans are the most essential participants in the real world, making their realistic simulation crucial. Thanks to the early success in generative models, there are some representative works [46], [103], [104] that introduce generative adversarial networks (GAN) [14] for animating

humans in videos. Despite these progress, the most essential problem in human video animation is the visual fidelity of the generated videos. ControlNet [105] and HumanSD [106] are plug-and-play methods for animating humans referring to poses based on a foundation text-to-image model, like Stable Diffusion [38]. Additionally, to resolve the generalization issue of these methods, animate-anyone [107] proposes a ReferenceNet to maintain more spatial details of the reference video and pushes generation quality in the wild to a new milestone. There are some follow-up works [108], [109] of animate-anyone trying to simplify the training architecture and costs. Besides, as the in-depth study of geometry and textures in the computer graphics area, there are some works that introduce 3D modeling into human video animation. Liquid Warping GAN [110], CustomHuman [111] and LatentMan [112] are early attempts introducing 3D human prior into the generation loop. The latest progress, MIMO [113], explicitly models the character, 3D motion, and scene respectively to drive human animation in the wild. These methods with or without 3D priors make a great step to introduce humans into the loop of the world simulator.

C. 3D Generation

3D generation focuses on both geometry and appearance to better simulate real-world scenarios. In this section, we explore various 3D representations and generation algorithms, providing a structured overview of recent advancements. Specifically, we categorize 3D generation methods based on their input modalities, including *Text-to-3D Generation*, which directly synthesizes 3D content from textual descriptions, *Image-to-3D Generation*, which introduces image constraints to refine text-driven outputs, and *Video-to-3D Generation*, which leverages video priors for more consistent 3D generation. A chronological summary of these advancements is presented in Figure 7, while Table IV offers a comprehensive comparison of cutting-edge methods. Notably, several approaches span multiple categories, demonstrating the versatility of modern 3D generation techniques.

Instead of constructing 3D generation models from scratch, most existing approaches are highly coupled with 2D and video generation models to leverage their powerful appearance modeling capability for better 3D generation, as illustrated in Figs. 5, 8, and 10. *First*, the image prior encoded in 2D and geometry cues encoded in video generation models can be employed to provide supervision for 3D generation models. *Second*, 2D and video generation models can be fine-tuned to take additional 3D information (e.g., normals) as input to synthesize 3D-aware multi-view images to facilitate 3D generation.

1) 3D Representation: In the domain of 3D generation, the choice of an optimal 3D representation is crucial. For neural scene representation, 3D data can be typically divided into three main categories: explicit, implicit, and hybrid representations, which are illustrated in Figure 4.

(1) Explicit Representations. Explicit representations offer precise visualizations of objects and scenes, defined by a set of elements. Traditional forms, such as point clouds, meshes, and voxels, have been widely used for years.

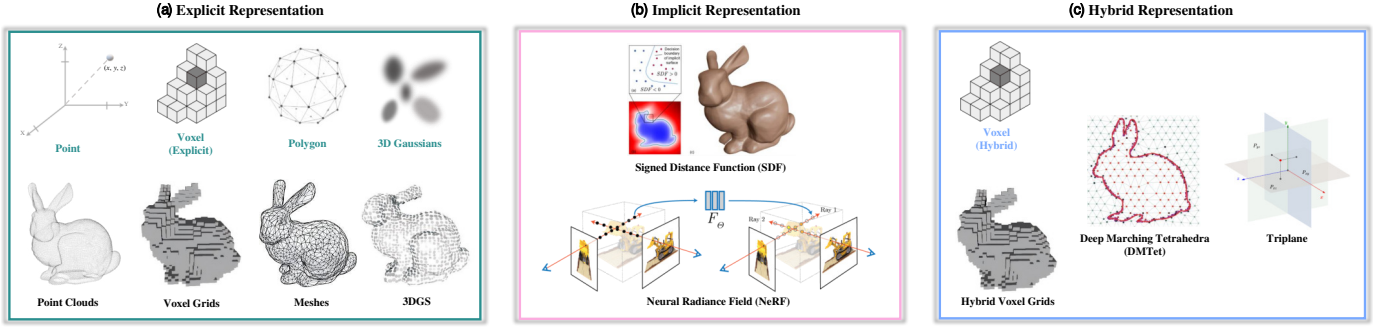


Fig. 4: Three main categories of neural scene representations. (a) Explicit representation stores geometry directly using point clouds, voxel grids [114], meshes [115], and 3D Gaussians [116]. (b) Implicit representation defines objects through functions like Signed Distance Functions (SDF) [117] and Neural Radiance Fields (NeRF) [118], enabling smooth, continuous surfaces without fixed resolution. (c) Hybrid representation combines explicit and implicit methods, using techniques like Hybrid Voxel Grids, Deep Marching Tetrahedra (DMTet) [119], and Triplanes for better efficiency and flexibility.

(i) **Point Clouds** are unordered sets of 3D points (x, y, z) and often enriched with attributes like color or normals. While widely used due to depth sensor output, their irregular structure [120], [121] challenges conventional neural networks in 2D domains.

(ii) **Voxel Grids** are composed of basic elements called voxels. A voxel, as a 3D counterpart of a pixel [122], represents points on a 3D grid. It subdivides a bounding box into smaller elements, each with an occupancy value. While voxels can store data like opacity or color, they are memory-inefficient for high-resolution data due to cubic memory growth ($\mathcal{O}(n^3)$) and sparse occupancy.

(iii) **Meshe**s composed of vertices and edges forming polygons (faces), define two-dimensional surfaces in three-dimensional space. They are more memory-efficient than voxel grids by encoding only object surfaces and, unlike point clouds, provide explicit connectivity, enabling geometric transformations and efficient texture encoding.

(iv) **3D Gaussian Splatting (3DGS)** is proposed as an effective method for accelerating both training and rendering tasks [116]. This technique models objects as collections of anisotropic Gaussian distributions, where each distribution is defined by its position (i.e., mean $\mathbf{x} \in \mathbb{R}^3$), covariance matrix Σ , opacity $\alpha \in \mathbb{R}$, and spherical harmonics coefficients $C \in \mathbb{R}^k$ (with k representing the degrees of freedom), allowing for the modeling of view-dependent color variations, which is expressed as:

$$G(\mathbf{x}) = e^{-\frac{1}{2}\mathbf{x}^T \Sigma^{-1} \mathbf{x}}. \quad (6)$$

To facilitate optimization, the covariance matrix Σ is typically factorized into a scaling matrix $\mathbf{S}$ and a rotation matrix $\mathbf{R}$, such that:

$$\Sigma = \mathbf{R} \mathbf{S} \mathbf{S}^T \mathbf{R}^T. \quad (7)$$

(2) Implicit Representations. Implicit representations describe 3D spaces with continuous functions, such as mathematical models or neural networks, capturing volumetric properties rather than surface geometry. Implicit neural representations approximate these functions with neural networks, enhancing expressiveness at the cost of higher training and inference overhead. Key approaches include Signed Distance Field (SDF) [117] and Neural Radiance Field (NeRF) [123].

(i) **Signed Distance Field (SDF)** is used to represent 3D shapes via neural fields, where a surface is implicitly defined as the zero-level set of the SDF. Given a point $\mathbf{x} \in \mathbb{R}^3$, the SDF function $f(\mathbf{x}) = d$ returns the shortest distance to the nearest surface point, with the sign indicating whether $\mathbf{x}$ lies inside ($d < 0$) or outside ($d > 0$) the shape. The surface is defined by the set of points where $f(\mathbf{x}) = 0$. The accuracy of SDFs relies on precise normal vectors, as their orientation determines distance signs, making accurate normal estimation essential for surface representation.

(ii) **Neural Radiance Field (NeRF)** [118], [123] represents 3D scenes as continuous volumetric functions encoded in the weights of a Multi-Layer Perceptron (MLP). Unlike DeepSDF [117], NeRF regresses density and color instead of a signed distance function, allowing for volumetric rendering. Specifically, NeRF maps spatial positions $\mathbf{x} \in \mathbb{R}^3$ and viewing directions $\mathbf{d} \in \mathbb{R}^2$ to density σ and color c , i.e., $f(\mathbf{x}, \mathbf{d}) = (\sigma, c)$, using an MLP trained on images with known camera poses. To render a pixel, NeRF casts a ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$ from the camera into the scene and samples points along the ray at intervals t_i . The final pixel color $C(\mathbf{r})$ is computed via volume rendering as:

$$C(\mathbf{r}) = \sum_i T_i \alpha_i c_i, \quad \text{where } T_i = \exp\left(-\sum_{k=0}^{i-1} \sigma_k \delta_k\right), \quad (8)$$

the term $\alpha_i = 1 - \exp(-\sigma_i \delta_i)$ here represents the opacity of the sampled point, and the accumulated transmittance T_i quantifies the probability that the ray travels from t_0 to t_i without being occluded by any particles. The term $\delta_i = t_i - t_{i-1}$ denotes the distance between consecutive sampled points along the ray.

(3) Hybrid Representations. Most current implicit methods depend on regressing NeRF or SDF values, which can limit their capacity to leverage explicit supervision on target views or surfaces. Explicit representations, however, offer useful constraints during training and improve user interaction. To leverage the complementary strengths of both paradigms, hybrid representations can be seen as a trade-off between explicit and implicit representations.

(i) **Hybrid Voxel Grids** can be used as hybrid representations in methods like [124]–[126]. [125] employs density and

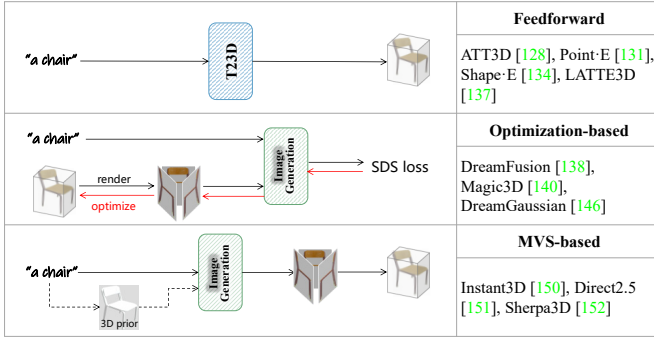


Fig. 5: Comparison between different text-to-3D generation paradigms. Image generation models serve as critical components to construct text-to-3D approaches.

feature grids for radiance field reconstruction, while Instant-NGP [126] uses hash-based multi-level grids, optimizing GPU performance for faster training and rendering.

(ii) **DMTet** [119] combines tetrahedral grids with implicit SDF for flexible 3D surface representation. A neural network predicts SDF values and position offsets for each vertex, allowing the modeling of complex topologies. The grids are converted into meshes via a differentiable Marching Tetrahedra (MT) layer, enabling efficient, high-resolution rendering. By optimizing geometry and topology with mesh-based losses, DMTet achieves finer details, fewer artifacts, and outperforms previous methods in conditional shape synthesis from coarse voxels on complex 3D datasets.

(iii) **Tri-plane** provides a memory-efficient alternative to voxel grids for 3D shape representation and neural rendering. It decomposes 3D volumes into three orthogonal 2D feature planes (XY, XZ, YZ). EG3D [127] utilizes this structure, employing an MLP to aggregate features from the planes and predict color and density values at any 3D point: $(\sigma, c) = \text{MLP}(\mathbf{f}_{xy}(\mathbf{x}) + \mathbf{f}_{xz}(\mathbf{x}) + \mathbf{f}_{yz}(\mathbf{x}))$. This approach reduces memory consumption compared to voxel-based NeRF and enables faster rendering.

2) **Algorithms: (1) Text-to-3D Generation.** To generate 3D content from the text prompt by simulating the geometries in the real world, extensive studies have been conducted and can be divided into three branches. Readers can refer to [128]–[130] for a more comprehensive survey in this field. Comparison of different branches of methods is illustrated in Fig. 5. As we can see, image generation models serve as critical components in text-to-3D approaches to provide supervision (i.e., SDS loss) or synthesize multi-view images for more accurate 3D generation.

(i) **Feedforward Approaches.** Motivated by text-to-image generation, a primary branch of methods extend existing successful generative models to directly synthesize 3D representations from the text prompt in one feedforward propagation. The key to success lies in encoding the 3D geometry into a compact representation and align it with the corresponding text prompt.

Michelangelo [131] first constructs a VAE model to encode 3D shapes into a latent embedding. Then, this embedding is

aligned with the ones extracted from language and image using a CLIP [30] model. Using a contrastive loss for optimization, a 3D shape can be inferred from the text prompt. ATT3D [132] uses Instant-NGP model as the 3D representation and bridges it with the text embedding using a mapping network. Then, multi-view images are rendered from the Instant-NGP model and the whole network is optimized using SDS loss. Motivated by ATT3D, Atom [133] learns to predict a triplane representation from the text embedding and employs a two-stage optimization strategy. Hyperfields [134] trains a dynamic hypernet to record the NeRF parameters learned from diverse scenes.

Recently, the impressive performance of diffusion models has motivated researchers to extend them to 3D generation. Early methods focus on learning to synthesize explicit 3D representations from the text prompt. Specifically, Point-E [135] first employs GLIDE [136] to synthesize multiple views, which are then used as conditions to produce a point cloud using a diffusion model. Later, MeshDiffusion [137] employ diffusions to build a mapping from text to meshes. Subsequent methods make attempts to applying diffusion models on implicit 3D representations. Shap-E [138] first maps 3D contents to parameters of a radiance field and then trains a diffusion model to generate these parameters conditioned on the text embedding. 3D-LDM [139] employs an SDF to represent the geometry of the 3D content and trains a diffusion model for text-conditioned generation. Similarly, Diffusion-SDF [140] constructs an SDF autoencoder with a voxelized diffusion model to generate voxelized signed distance fields (SDFs) from the text prompt. LATTE3D [141] develops a texture network and a geometry network to generate NeRF and SDF conditioned on the text embedding, respectively. Then, a 3D-aware diffusion model is optimized through an SDS loss.

Discussion. Compared with optimization-based approaches, feedforward approaches favor high efficiency and is capable of generating 3D contents without test-time optimization. These approaches, however, rely heavily on the quantity of data and usually suffer inferior structure and texture details.

(ii) **Optimization-based Approaches.** On top of text-to-image generation, another branch of methods optimize 3D representations by leveraging powerful text-to-image generative models to provide rich supervision.

DreamFusion [142] first introduces score distillation sampling (SDS) loss to optimize NeRFs using images synthesized from text prompts. MVDream [143] fine-tunes a multi-view diffusion model to generate multi-view images with cross-view consistency to train a NeRF to capture 3D contents. Magic3D [144] employs textured meshes to represent the 3D object and adopts SDS loss for optimization of the meshes. Dream3D [145] first generates an image from the text prompt, which is then employed to produce the 3D shape to initialize a neural radiance field. Next, a NeRF is optimized using the CLIP guidance. Fantasia3D [146] further combines DMTet and SDS loss to generate a 3D object from the text prompt. ProlificDreamer [147] develops variational score distillation (VSD) to model the distribution of 3D representations and produces higher-quality results with rich details. To address the multi-face Janus issue, PI3D [148] first fine-tunes the text-to-

	Time ($\downarrow$)	Quality ($\uparrow$)	Alignment ($\uparrow$)
Magic3D [144]	40min	38.7	35.3
Fantasia3D [146]	45min	29.2	23.5
DreamFusion [142]	30min	24.9	24.0
ProlificDreamer [147]	240min	51.1	47.8
MVDream [143]	30min	53.2	42.3
DreamGaussian [150]	7min	19.9	19.8
VP3D [149]	-	54.8	52.2
GaussianDreamer [157]	-	54.0	-

TABLE III: Quantitative comparison between different text-to-3D methods. Results are obtained from [157].

image diffusion model to produce pseudo-images. Then, these images are adopted to generate a 3D shape using an SDS loss. VP3D [149] first uses a text-to-image diffusion model to generate a high-quality image from the text prompt. Then, 3D representations are optimized via SDS loss using the resultant image and the text prompt as conditions.

With remarkable advances in 3D Gaussian, it is widely studied in the field of text-to-3D generation. DreamGaussian [150] first employs a diffusion to obtain 3D Gaussians with SDS loss being used for optimization. Then, meshes are extracted from the 3D Gaussians with texture being refind to obtain higher quality contents. To promote convergence, GSGEN [151] and GaussianDreamer [152] first employ Point-E to generate a point cloud from the text prompt to initialize the positions of Gaussians. Then, these Gaussians are optimized to refine their geometry and appearance using SDS loss. Sculpt3D [153] introduces a 3D prior by retrieving reference 3D objects in a database, which can be seamlessly integrated with existing pipelines.

Discussion. Thanks to the rich knowledge in text-to-image models, optimization-based approaches produce finer details. However, these approaches require expensive per-prompt optimization and are time-consuming.

(iii) MVS-based Approaches. Instead of directly generating 3D representations from text prompts, to make better use of text-to-image models, numerous attempts have been made to synthesize multi-view images for 3D generation.

Instant3D [154] first fine-tunes text-to-image diffusion model to generate four-view images. Then, these images are passed to a transformer to predict triplane representation. Direct2.5 [155] fine-tunes a multi-view normal diffusion model on 2.5D rendered and natural images. Given a text prompt, Direct2.5 first produces normal maps and optimizes them through differentiable rasterization. Then, the optimal normal maps are adopted as conditions to synthesize multi-view images. Sherpa3D [156] first employs a 3D diffusion model to generate a coarse 3D prior from the text prompt. Then, the normal map is produced and used to synthesize multi-view images with 3D coherence.

Discussion. With recent advances of VLMs, lifting these 2D generative models for 3D generation by injecting 3D priors has drawn increasing interests. However, the formulation of 3D consistency and fine-tuning with limited amount of 3D data still remain open problems.

Evaluation. Quantitative evaluation of text-to-3D methods remains an open problem. For subjective quality evaluation, common reference-based metrics (e.g., PSNR) are not appli-



Fig. 6: Qualitative comparison between different text-to-3D methods.

cable as groundtruth data are unavailable while non-reference quality (e.g., FID) metrics may not always align with human preferences. Consequently, most methods employ CLIP Score and CLIP R-Precision to evaluate the alignment of 3D models with text prompts. Recently, several benchmarks [158], [159] are established to comprehensively assess text-to-3D generation methods. Here, we report the quantitative scores of representative methods in Table III and present their visual results in Fig. 6. Readers can refer to [158], [159] for more details.

(2) Image-to-3D Generation. The goal of image-to-3D task is to generate high-quality 3D assets that are consistent with the identity of the given image. Due to the high cost of 3D data collection, text-to-3D generation lacks sufficient high-quality text annotations to scale up compared with image and video generation. Since images naturally capture more low-level information that closely align with the 3D modality, the image-to-3D task narrows the modality gap between the input and the output compared to text-to-3D generation. Consequently, image-to-3D has emerged as a foundational task for advancing native 3D generation. To leverage the knowledge inside image generation models, they are frequently adopted as components of image-to-3D models (Fig. 8). The qualitative comparison of the part methods is shown in Fig. 9 and the quantitative comparison is illustrated in Table V. Due to inconsistencies in the evaluation datasets or metrics used in the paper, some works are not listed in the figure and table.

(i) Feedforward Approaches. These methods first encode 3D assets into latent codes with a compression network, like VAE, and then train a generative model for the latent space samples. 3DGen [160] introduce triplane as latent space, improving both the accuracy and efficiency of the compression network. Direct3D [163] adopt the triplane representation and directly use 3D supervision for training, preserving detailed 3D information in the latent triplane. Michelangelo [131] draws inspiration from 3Dshape2vecset [184], and utilize 1D vectors as the latent space, when supervise the output with occupancy fields. CraftsMan [162] further introduces multi-view generation model to generate multi-view images as conditions of the diffusion model, followed with normal-based refinement for generated meshes. Clay [161] brings a comprehensive system pretrained on large-scale 3D datasets for 3D generation, including VAE and diffusion model based on 1D vectors for geometry generation, Material Diffusion for PBR

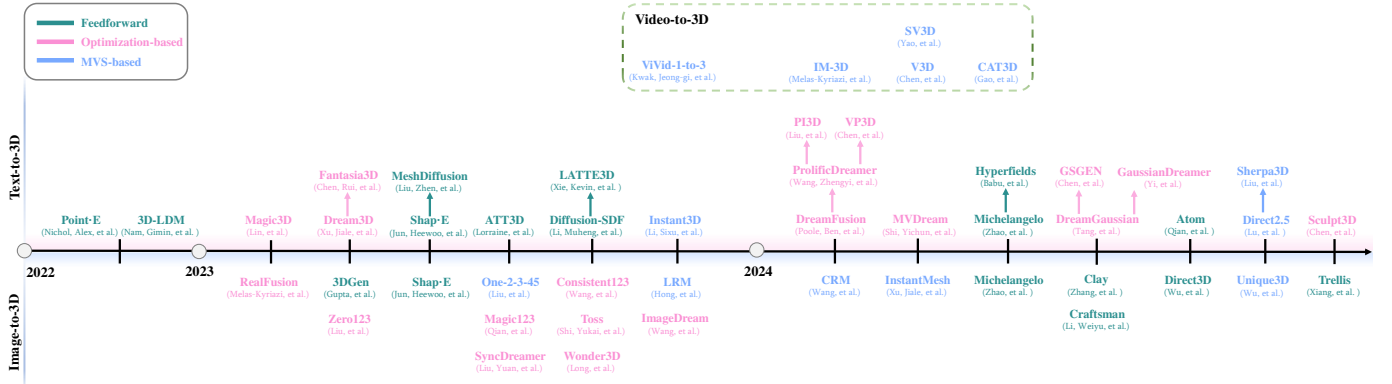


Fig. 7: Chronological overview of recent text-to-3D, image-to-3D and video-to-3D generation methods.

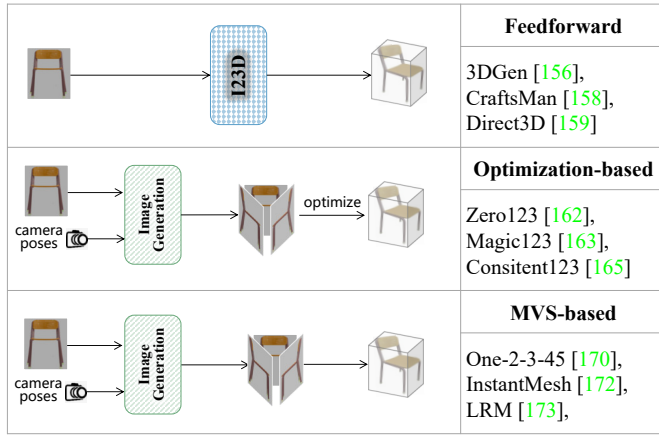


Fig. 8: Comparison between different image-to-3D generation paradigms. Image generation models can be employed to synthesize multi-view images to promote 3D generation.

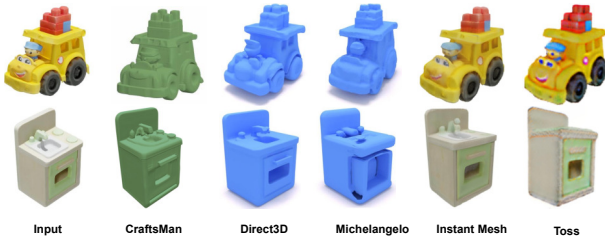


Fig. 9: Qualitative comparison between different image-to-3D methods.

texture, and conditioning design across various modalities.

Discussion. Native methods train a compression network and generation model on 3D datasets, and demonstrate superior performance in geometry generation compared to the MVS-based and optimization-based methods, capable of producing more fine-grained geometry details. However, the size of 3D datasets [185], [186] scales much slower than image or video datasets [187], [188] due to the high cost of production and collection. Therefore, native methods lack sufficiently diverse and extensive data for pretraining. Consequently, how to leverage the priors from videos and images to enhance the diversity and generalization of 3D generation, especially in

texture generation, remains an area for further exploration.

(ii) Optimization-based Approaches. With the development of distillation-based methods in text-to-3D models, optimization-based methods directly optimize 3D assets through a training process supervised by SDS loss from pretrained image-to-image or text-to-image generation models, when maintaining the image identity by various additional loss constraints.

Adapted from Dreamfusion [142], Magic3D [144] and SJC [189], RealFusion [165] only extracts priors from pretrained text-to-image models with the SDS loss, when the image reconstruction loss and textual inversion is used to preserve low-level and semantic identity respectively. With the advent of large-scale open-set 3D datasets [185], Zero123 [166] introduces the priors of 3D datasets into image-to-3D tasks by replacing the text-to-image model with a novel view synthesis model for distillation. Specifically, Zero123 finetunes a pretrained image-image generation model on the 3D datasets by introducing camera poses as a condition to control the generated image’s viewpoint. The pretrained novel view synthesis model preserves both the details from the image models and the multi-view consistency from 3D datasets, which significantly mitigates the Janus problem.

A series of works have expanded upon Zero123 [166]. Zero123-xl [186] pretrains Zero123 pipelines on a $10\times$ larger 3D datasets for better generalization. Magic123 [167] leverages 2D and 3D priors simultaneously for distillation to manage the trade-off between generalization and consistency, and uses coarse-fine pipeline for higher quality. SyncDreamer [168] and Consistent123 [169] both further improves the multi-view consistency of NVS models by introducing a synchronized multi-view diffusion model, when the former leverages 3D volume to model joint distribution relations of images and the latter utilizes cross-view attention and shared self-attention. Toss [170] additionally brings text caption as high-level semantics of 3D data into the NVS model pretraining for stronger plausibility and controllability of invisible views. ImageDream [171] addresses both multi-view consistency and 3D details problem by designing a multi-level image-prompt controller and training with text descriptions. Wonder3D [173] incorporates the cross-domain attention mechanism, enabling the NVS model to simultaneously denoise images and align normal maps, when introducing normal maps into the optimization

TABLE IV: Recent text-to-3D, image-to-3D and video-to-3D generation methods.

Paradigms	Methods	Paper	Code	Representations	Objectives
Text-to-3D	Feedforward	Point-E [135]	Link	Point Clouds	MAE
		3D-LDM [139]	Link	SDF	reconstruction+regularization
		ATT3D [132]	Link	Instant-NGP model	SDS
		Diffusion-SDF [140]	Link	SDF	reconstruction+KL-Divergence
		LATTE3D [141]	Link	NeRF/SDF	SDS+regularization
		MeshDiffusion [137]	Link	Mesh	diffusion loss
		Shap-E [138]	Link	NeRF	rendering loss
		Hyperfields [134]	Link	NeRF	SDS
		Michelangelo [131]	Link	Occupancy Field	text-image-shape contrastive + BCE
		Atom [133]	Link	Triplane	SDS
	Optimization	Magic3D [144]	Link	NeRF/DMTet	SDS
		Dream3D [145]	Link	NeRF	reconstruction+regularization
		Fantasia3D [146]	Link	DMTet	SDS
		DreamFusion [142]	Link	NeRF	SDS
		ProlificDreamer [147]	Link	NeRF	VSD
		PI3D [148]	Link	Triplane	SDS
		VP3D [149]	Link	NeRF	SDS
		MVDream [143]	Link	NeRF	SDS
		DreamGaussian [150]	Link	3DGS	SDS
		GaussianDreamer [157]	Link	3DGS	SDS
Image-to-3D	Feedforward	GSGEN [151]	Link	3DGS	SDS
		Sculpt3D [153]	Link	NeRF	diffusion loss
		Instant3D [154]	Link	Triplane	MSE+LPIPS
		Direct2.5 [155]	Link	Multi-view normal maps	normal rendering+alpha mask loss
		Sherpa3D [156]	Link	DMTet	SDS
	Optimization	3DGen [160]	Link	DMTet	rendering loss
		Shap-E [138]	Link	NeRF	rendering loss
		Michelangelo [131]	Link	Occupancy Field	text-image-shape contrastive + BCE
		Clay [161]	Link	Occupancy Field	BCE
		CraftsMan [162]	Link	Occupancy Field	BCE
		Direct3D [163]	Link	Occupancy Field	BCE
		Trellis [164]	Link	3DGS/RF/Mesh	MSE
		RealFusion [165]	Link	NeRF	SDS/Image Reconstruction
		Zero123 [166]	Link	NeRF	SDS
		Magic123 [167]	Link	NeRF	SDS
Video-to-3D	MVS	SyncDreamer [168]	Link	NeRF	SDS
		Consistent123 [169]	Link	NeRF	SDS
		Toss [170]	Link	NeRF	SDS
		ImageDream [171]	Link	NeRF	SDS
		IPDreamer [172]	Link	NeRF	SDS
		Wonder3D [173]	Link	Mesh	Image/Normal MSE
		One-2-3-45 [174]	Link	SDF	Image/Depth MSE
		CRM [175]	Link	Triplane	Image/Depth MSE
		InstantMesh [176]	Link	Triplane	Image/Depth/Normal MSE
		LRM [177]	Link	NeRF	MSE
Video-to-3D	MVS	Unique3D [178]	Link	Mesh	Image/Normal MSE
		ViVid-1-to-3 [101]	Link	Multi-view images	-
		IM-3D [179]	Link	3DGS	MSE+LPIPS
		V3D [180]	Link	Mesh	MSE+LPIPS
		SV3D [181]	Link	NeRF / DMTet	MSE+LPIPS
Video-to-3D	MVS	CAT3D [182]	Link	NeRF	MSE+LPIPS

TABLE V: Quantitative comparison with baseline methods on the GSO dataset [183].

Type	Method	CD↓	IoU↑	Time↓
Optimization-based Approaches	Realfusion [165]	0.0819	0.2741	~90min
	Magic123 [167]	0.0516	0.4528	~60min
MVS-based Approaches	One-2-3-45 [174]	0.0629	0.4086	~45s
	Zero123 [166]	0.0339	0.5035	~10min
	InstantMesh [176]	0.0187	0.6353	~10s
Feedforward Approaches	Point-E [135]	0.0426	0.2875	~40s
	Shap-E [138]	0.0436	0.3584	~10s
	Michelangelo [131]	0.0404	0.4002	~3s
	CraftsMan3D [162]	0.0291	0.5347	~5s

process additionally. IPDreamer [172] enables controllable 3D synthesis from complex image prompts by introducing IPSDS, a variant of SDS, and a mask-guided alignment strategy for multi-prompt consistency.

Discussion. Inheriting the powerful priors of image generation models, optimization-based methods demonstrate strong generalization capabilities and can model high-precision tex-

tures. However, since novel view synthesis (NVS) models only use 2D data sampled from 3D rather than directly 3D data for supervision during pretraining, the multi-view consistency problem cannot be fundamentally resolved, despite improvements through 3D volume modeling or cross-view attention. Consequently, optimization-based methods often suffer from overly smooth geometries and long times for training due to the optimization paradigm.

(iii) MVS-based Approaches. MVS-based methods split the image-to-3D generation into two stages: first generate multi-view images from a single image using an NVS model, then directly create 3D assets from these multi-view images using a feedforward reconstruction network.

Based on the multi-view images predicted by Zero123 [166], One-2-3-45 [174] proposes an elevation estimation module and utilizes an SDF-based generalizable neural surface reconstruction module pretrained on 3D datasets for 360° mesh reconstruction, which reduces the reconstruction time to 45 seconds compared to optimization-based. CRM [175]

further freeze the output images of the multi-view generation model to six fixed camera poses, largely improving the consistency between multi-views. Then CRM input multi-view images into a convolutional U-Net to create a high-resolution triplane supervised by depth and RGB images. InstantMesh [176] also freeze the camera poses of multi-view images, but employ a transformer-based multi-view reconstruction model based on LRM [177] to reconstruct 3D meshes, providing better generalization at the expense of some image-to-3D detail consistency. Unique3d [178] introduce a multi-level upscale strategy to progressively generate higher-resolution multi-view images, and a normal map diffusion model to predict multi-view normal maps for the initialization of coarse meshes, which is refined and colorized based on multi-view images.

Discussion. Compared with optimization-based methods, MVS-based methods train a feedforward reconstruction model on 3D datasets to reconstruction a high-quality 3D from multi-view images, significantly improving the 3D consistency and reducing inference time to seconds level. However, MVS-based methods often lack high-quality geometry details due to the limitation of the model scale.

(3) Video-to-3D Generation. Video-to-3D methods fundamentally build upon advances in 2D diffusion models, extending image-based generative priors into the temporal domain. By modeling coherent sequences of frames—with consistent texture, lighting, and geometry—video diffusion enables implicit 3D structure learning through motion and viewpoint variation captured in 2D. Frameworks like SV3D, Hi3D, and V3D adapt pretrained video diffusion backbones to generate multi-view frame sequences, which are then integrated via volume rendering, mesh optimization or Gaussian splatting to reconstruct explicit 3D geometry. This synergy leverages the strong content synthesis capabilities of 2D video models while introducing spatial consistency and camera control essential for dense 3D reconstruction. The vast corpus of online video data serves as a rich repository of 3D information, capturing object motions, viewpoint variations, and camera transitions that unveil multi-view perspectives often unavailable in static imagery [190]–[195]. This dynamic content provides both temporal coherence and spatial consistency across sequential frames, which are essential for understanding complex 3D scenes and generating high-fidelity 3D structures [13]. As a result, leveraging these multi-view and time-variant data has become a promising approach for reconstructing and synthesizing 3D-consistent objects [196]. Recent research explores video-based priors for robust 3D generation [180]–[182], aiming to learn 3D representations that maintain coherence across frames and adapt to changing viewpoints. At high-level, the main idea of these works for video-to-3D generation is to enable a camera-controllable video model as a consistent multi-view generator for dense 3D reconstruction (Fig. 10).

Recent advancements in video diffusion models have illuminated their exceptional capabilities in generating realistic videos while providing implicit reasoning about 3D structures. However, significant challenges persist in employing these models for effective 3D generation, particularly regarding precise camera control. Traditional models [197]–[199] are

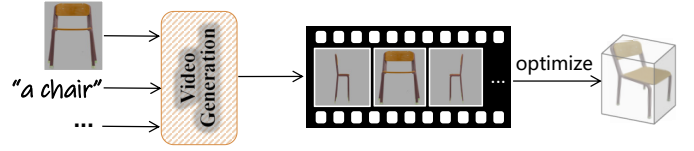


Fig. 10: An illustration of video-to-3D generation paradigm. Camera-controllable video models can be used as a consistent multi-view generator for 3D generation.

typically confined to generating clips with smooth and short camera trajectories, limiting their ability to create dynamic 3D scenes or integrate varying camera angles effectively. To address these limitations, several innovative techniques have been developed to enhance camera control within video diffusion frameworks. One early approach is AnimateDiff [58], which employs Low-Rank Adaptation (LoRA) [200] to fine-tune video diffusion models with fixed camera motion types. This method facilitates the synthesis of structured scenes while adhering to specified camera dynamics. Another significant advancement is MotionCtrl [201], which introduces conditioning mechanisms that allow models to follow arbitrary camera paths, thereby enabling greater flexibility in generating diverse perspectives and overcoming the rigidity of previous methods. Based on the availability of camera-controllable video generation, SVD-MV [202], SV3D [181], and IM-3D [179] explore how to leverage camera control to improve the generation of 3D objects from video data. For example, SV3D trains a video diffusion model capable of rendering arbitrary views, which demonstrates enhanced generalization and high-resolution output (576×576 pixels). Such capability allows for maintaining spatial coherence across frames while adapting to various viewpoints, effectively addressing critical challenges in dense reconstruction. While effective, these methods often restrict camera movements to fixed, orbital paths surrounding central objects, which limits their applicability to complex scenes with rich contextual backgrounds. However, many of these methods still fall short when producing compelling 3D representations of intricate environments, where varying camera angles and interactions with multiple objects are crucial.

As controlling camera movements in video models complement novel view information, several methods have explored the potential of video diffusion models for novel view synthesis (NVS). For example, Vivid-1-to-3 [101] effectively merges a view-conditioned diffusion model with a video diffusion model, allowing for the generation of temporally consistent views. By ensuring smooth transitions across frames, this model enhances the quality of the synthesized output, making it particularly effective for 3D scene representations. CAT3D [182] augments rich multi-view information with a multi-view diffusion model.

Discussion. Leveraging video priors in multi-view generation transforms video diffusion models into consistent multi-view generators for dense 3D reconstruction. Further exploration will enhance high-fidelity 3D representation, especially for complex, dynamic environments requiring robust multi-view synthesis.

3) *Applications: (1) Avatar Generation.* With the emergence of the metaverse and the popularity of VR/AR, 3D avatar generation has drawn increasing interests. Early works focus on generating head avatars [203]–[205], which leverage text-to-image diffusion models and neural radiance fields to create facial assets. Later methods pay more attention to realistic full-body avatars generation by integrating neural radiance fields with statistical models [206], [207]. Recently, the animation capability of avatar generation has gained great attention with numerous methods being proposed [208], [209].

(2) *Scene Generation.* In addition to avatar generation, scene generation to create a realistic 3D environment is also highly demanded in applications like metaverse and embodied intelligence. Early methods focus on object-centered scenes and leverage conditional diffusion models to synthesize multi-view images to optimize neural radiance fields [139], [140]. Later works extend these methods to room-scale scenes by introducing progressive strategies [210], [211]. Motivated by their success, recent studies further investigate the generation of outdoor scenes, ranging from street-scale [212], [213] to city-scale [214], [215].

(3) *3D Editing.* The powerful 3D generation capability has created the downstream application of 3D content editing. Several methods focus on changing the appearance or geometry of the 3D content globally [216], [217] without isolating a specific region from a scene. For example, scene stylization methods [218], [219] aims to manipulate the style of a 3D asset, such as illumination tuning and climate changes. Recent efforts are made to achieve flexible 3D content editing at a finer-grained level. Specifically, appearance change [220], [221], geometry deformation [222], [223], and object-level manipulation [224], [225] have been studied, with promising editing results being achieved.

D. 4D Generation

We culminate in 4D generation by integrating all dimensions. As a cutting-edge field in computer vision, 4D generation focuses on synthesizing dynamic 3D scenes that evolve temporally based on multimodal inputs such as text, images, or videos. Unlike traditional 2D or 3D generation [226], 4D synthesis introduces unique challenges, requiring both spatial coherence and temporal consistency while balancing high fidelity, computational efficiency, and dynamic realism [227]. In this section, we first introduce 4D representation, which builds upon 3D representation, and then summarize current 4D generation methods. Recent research has explored two primary paradigms: optimization-based methods leveraging Score Distillation Sampling (SDS) and feedforward-based approaches that avoid per-prompt optimization. These paradigms address distinct technical challenges, highlighting the field’s complexity and the ongoing pursuit of a feasible balance among visual quality, computational efficiency, and scene flexibility. Representative works in 4D generation are summarized in Table VI.

1) *4D Representation:* The field of 4D representation, which incorporates a temporal dimension into 3D modeling, provides a strong foundation for understanding dynamic

scenes. By extending static 3D spatial representations (x, y, z) with time (t) , these methods encode scene dynamics and transformations, essential for applications such as non-rigid human motion capture and simulating object trajectories [257]–[260]. Most 4D representations can be factorized into canonical 3D representation and deformation. The first module canonical 3D representation is used to model a static template shape. The second module deformation is used to deform the template shape to synthesize motion. To deform the canonical 3D representation, here are some common used deformation representations including deformation fields that is a neural networks that map a space-time point to its corresponding position on the canonical template shape and deformation primitives like linear blend skinning which represents the motion of a point through a combination of rigid motions associated with different body parts or control points. There are pros and cons to these two types of deformation representations. Deformation fields are more flexible than deformation primitives, which means deformation fields are more general and can theoretically fit complex motions. Deformation primitives designed for articulated objects such as human or animal motions, however, have the advantage of robustness for large articulated motion since deformation fields lack the inductive bias needed for accurate reconstruction, especially with rapid motion. Below, we focus most on the 3D canonical representation with deformation fields type of 4D representation.

A major challenge in 4D representation is the high computational cost of reconstructing a single scene. To address this, explicit and hybrid methods enhance efficiency without sacrificing quality. For instance, planar decompositions streamline 4D spacetime grids by breaking them into smaller components [261]–[263], while hash-based representations reduce memory and processing requirements [264]. 3DGS balances speed and quality by using deformation networks to adapt static Gaussians to dynamic ones [116], [265].

Recent advances disentangle static and dynamic scene components to render both rigid and non-rigid motions efficiently. For example, D-NeRF first encodes scenes into a canonical space and then maps them into temporally deformed states [266]. 3D Cinemagraphy generates feature-based point clouds from single images and animates using 3D scene flow [267]. 4DGS captures temporal dynamics by modeling attributes like scales, positions, and rotations as time functions while keeping static scene unchanged [268].

Hybrid NeRF-based methods expand 4D modeling with plane- and voxel-based feature grids. These grids, combined with MLPs, enable efficient novel-view synthesis and extend to dynamic scenes by incorporating temporal planes [261], [262]. Deformable NeRFs separate geometry and motion, simplifying motion learning and supporting applications like image-to-4D video generation and multi-view reconstruction [126]. Collectively, these advancements reflect ongoing progress in achieving computationally efficient, high-quality temporal modeling for dynamic scenes.

2) *Algorithms:* Modern 4D generation methods are largely rooted in the foundations laid by 3D generation. In particular, advances in 3D reconstruction, such as NeRF and 3DGS, have directly influenced how we model and render dynamic

TABLE VI: Representative works of 4D generation methods. "Rep" stands for representations.

Approaches	Methods	Paper	Code	Representations	Priors / Models	Objectives
Feedforward	Control4D [228]	Link	-	GaussianPlanes	GAN-based	GAN objective
	Animate3D [229]	Link	Link	4DGS	Diffusion-based	latent diffusion loss
	Vidu4D [230]	Link	Link	Dynamic Gaussian Surfels	Diffusion-based	reconstruction/regularization loss
	Diffusion4D [231]	Link	Link	4DGS	Diffusion-based	latent diffusion + motion reconstruction loss
	L4GM [232]	Link	Link	3DGS	VAE-based	MSE+LPIPS
	GenXD [233]	Link	Link	4DGS	Diffusion-based	SSIM+LPIPS
	CAT4D [234]	Link	-	Deformable 3DGS	Diffusion-based	LPIPS
	4Real-Video [235]	Link	-	Deformable 3DGS	Diffusion-based	velocity matching loss of rectified flow [236]
	MAV3D [237]	Link	-	HexPlane-/NeRF-based 4D Rep	T2I/T2V	SDS
	4D-fy [238]	Link	Link	Hash-/NeRF-based 4D Rep	MVDream/T2I/T2V	Hybrid SDS
Optimization	AYG [239]	Link	-	Deformable 3DGS	MVDream, T2I/T2V	SDS variant(CSD [240])
	Dream-in-4D [241]	Link	Link	NeRF-based 4D Rep	MVDream/T2I/T2V	Hybrid SDS
	TC4D [242]	Link	Link	Hash-/NeRF-based 4D Rep	T2V	SDS
	4Real [243]	Link	-	Deformable 3DGS	T2V	SDS
	C3V [244]	Link	-	3DGS	LDMs [38]	SDS
	Consistent4D [245]	Link	Link	D-NeRF	Zero123	SDS
	SC4D [246]	Link	Link	SC-GS [247]	SVD/Zero123	SDS
	STAG4D [248]	Link	Link	Deformable 3DGS	SVD/Zero123	SDS
	DreamScene4D [249]	Link	Link	Deformable 3DGS	SVD/Zero123	SDS
	4DM [250]	Link	Link	D-NeRF	SVD	4D-aware SDS + anchor loss
	DreamMesh4D [251]	Link	Link	SC-GS+Mesh Rep	Zero123	SDS
	4K4DGen [252]	Link	Link	4DGS	SVD	L1
	EG4D [253]	Link	Link	4DGS	SVD	L1
	AvatarGO [254]	Link	Link	3DGS+Hexplane	SVD/Zero123	SDS
	Disco4D [255]	Link	Link	4DGS	SVD	SDS
	GenMOJO [256]	Link	Link	Deformable 3DGS	SVD/Zero123	SDS

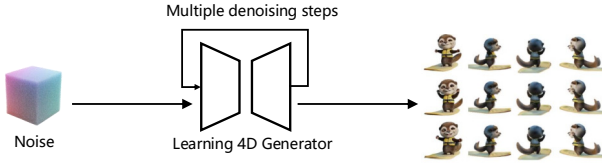


Fig. 11: An illustration of 4D generation feedforward paradigm. We directly train the 4D generator on the multi-view dynamic data.

4D scenes. These 3D frameworks provide not only efficient data structures and rendering techniques but also critical inductive biases that benefit temporal modeling in 4D. At the representation level, 3D methods offer canonical spatial priors that can be extended with deformation fields or motion trajectories to capture temporal evolution. From a training perspective, fast-training techniques from 3D (e.g., hash encoding, hierarchical sampling) have been adapted to accelerate 4D optimization. Human animation, as a most representative 4D task, particularly benefits from foundations and advancements in 3D human modeling. Techniques such as SMPL(-X), linear blend skinning, and neural deformation fields provide strong structural priors for modeling articulated motion. These 3D-based tools enhance both the realism and controllability of 4D human motion synthesis.

(1) Feedforward Approaches. Feedforward-based methods provide an efficient alternative by generating 4D content in a single forward pass (Fig. 11), bypassing the iterative optimization required in SDS-based pipelines. These methods rely on pre-trained models, leveraging temporal and spatial priors to achieve fast and consistent generation. Control4D [228] and Animate3D [229] directly synthesize dynamic scenes from textual or visual inputs, enabling real-time applications such as interactive media and personalized content creation. Vidu4D [230] improves motion trajectories by incorporating temporal priors, ensuring frame-to-frame coherence and

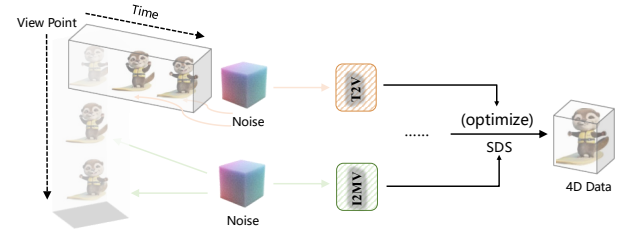


Fig. 12: An illustration of 4D generation optimization-based paradigm. We use SDS guidance to make the optimized 4D asset follow the pretrained text-to-image, multi-view image, and text-to-video prior.

smooth transitions. Diffusion4D [231] extends the capabilities of diffusion models to handle 4D scene synthesis by combining spatial-temporal feature extraction with efficient inference mechanisms. L4GM [232] further enhances feedforward techniques by integrating latent geometry modeling, producing high-quality results while maintaining computational efficiency.

Discussion. Feedforward-based approaches excel in scenarios that prioritize speed and adaptability, such as real-time content generation and lightweight deployment on consumer devices. However, their reliance on pre-trained models and limited flexibility in handling complex dynamics pose challenges in achieving the same level of detail and diversity as optimization-based methods. Despite these limitations, feedforward techniques represent a significant step toward practical 4D generation, addressing key challenges of computational efficiency and scalability. By bridging the gap between quality and speed, these methods are poised to play a critical role in advancing 4D content generation for a wide range of applications.

(2) Optimization-based Approaches. Optimization-based methods are foundational in 4D generation, employing iterative techniques such as Score Distillation Sampling (SDS)

	Time (↓)	CLIP (↑)	LPIPS (↓)
Consistent4D [245]	2hr	0.87	0.16
4DGen [269]	1hr	0.89	0.13
STAG4D [248]	1hr	0.91	0.13
DG4D [270]	10min	0.87	0.16

TABLE VII: Quantitative comparison between different 4D generation methods. Results are obtained from [232].

to adapt pre-trained diffusion models to synthesize dynamic 4D scenes (Fig. 12). These approaches leverage powerful priors from text-to-image, multi-view image, and text-to-video generation models, achieving temporally coherent scenes with rich motion dynamics. For example, MAV3D [237] optimizes NeRF or HexPlane features against SDS loss guided by textual prompts, while 4D-fy [238] and Dream-in-4D [241] improve the 3D consistency and motion dynamics by integrating image, multi-view, and video diffusion models in SDS supervision. AYG [239] proposes to use deformable 3DGS as inherent representation, which can easily disentangle static geometry from dynamic motion with a simple delta deformation field to improve flexibility.

Based on such pipelines, recent works further improve 4D generation from multiple aspects: appearance quality, geometry consistency, motion faithfulness, and generation controllability. In particular, TC4D [242] and SC4D [246] enable free user control on 4D object motion trajectory. STAG4D [248] employs multi-view fusion to enhance spatial and temporal alignment across frames, ensuring smooth transitions and consistency. Additionally, DreamScene4D [249] and DreamMesh4D [251] adopt disentanglement strategies to localize optimization efforts, significantly reducing computational overhead while maintaining high fidelity. Recent advances such as 4Real [243] and C3V [244] further push the boundaries of optimization-based methods by combining compositional scene generation with efficient optimization. These methods break dynamic scenes into modular components such as static geometry and motion fields, enabling flexible updates and diverse content generation. Despite their strengths in achieving high-quality and temporally consistent results, optimization-based approaches remain computationally demanding, with runtime requirements often precluding real-time applications. As research progresses, ongoing efforts focus on improving scalability and reducing latency without compromising visual fidelity or dynamic realism.

Evaluation. Quantitative evaluation of 4D generation methods mainly focuses on three aspects: 1) Time cost to generate a single 4D asset. 2) CLIP score, which evaluates on image similarity between the generated and the ground truth novel views. 3) Perceptual similarity (LPIPS) between the generated and the ground truth novel views. Here, we report the quantitative scores of representative methods in Table VII and present their visual results in Fig. 13. Readers can refer to [232] for more details.

3) *Applications:* (1) **4D Editing.** Instruction-guided editing, which allows users to edit scenes through natural language, offers a user-friendly and intuitive approach. While this has been successfully achieved for 2D images with models

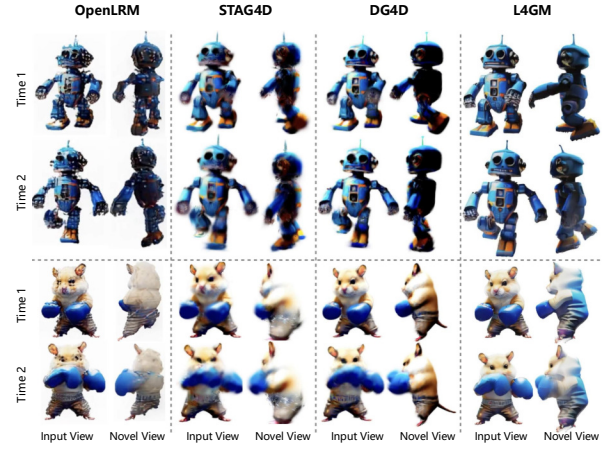


Fig. 13: Qualitative comparison between different 4D generation methods. Results are obtained from L4GM [232].

like Instruct-Pix2Pix (IP2P) [271] and Instruct-NeRF2NeRF (IN2N) [217] for 3D scenes, extending this capability to 4D scenes presents significant challenges. Recent developments in text-to-image diffusion models and differentiable scene representations have enabled editing 4D scenes using text prompts. For instance, Instruct 4D-to-4D [272] treats 4D scenes as pseudo-3D scenes, adopting a video editing approach to iteratively generate coherent edited datasets. Concurrent work like Control4D [228] employs GANs alongside diffusion models to ensure consistent editing for dynamic 4D portraits based on text instructions.

(2) **Human Animation.** As an essential component of 4D simulation, human motion generation is a research branch that attracts the most attention from the community. Different from the human-centric video generation (in Sec. III-B2), 3D human motion generation is easier to animate characters in 3D applications, *e.g.* games and embodied intelligence. The recent success of the 3D human motion generation mainly comes from the well-studied human body parameterized models [273], [274].

The human motion generation aims to achieve a basic target: simulating 4D human subjects in the digital world, which can be divided into two folds. 1) *Generating motions with sparse control signals* mainly simulates human animations in the virtual world based on the sparse motions specified by users. Robust motion in-between [275] proposes a time-to-arrival embedding and the scheduled target noise vector to perform motion in-between with different transition lengths in a model robustly. As the phase manifold of the motion space has a good structure, Starke *et al.* [276] propose a mixture-of-expert network to in-between motions in the phase manifold. Besides, another part of sparse control-guided motion generation is motion prediction, *a.k.a* motion extrapolation. Early researches in motion prediction [277]–[280] try to predict motions in a deterministic way. Considering that motion prediction is subjective, several works [281]–[285] have proposed generating diverse motions in their predictions. 2) *Generating motions with multi-modality conditions* aims to simulate human motions with other modality inputs, *e.g.* text, audio, musics. To

resolve the problem of limited pairwise text-motion data, Guo *et al.* [286] propose a relatively large text-motion dataset, HumanML3D, which is significantly larger than previous datasets and greatly pushes the development of the tasks. Besides, some researchers verify that VQ-VAE [286]–[291] is another effective fashion for synthesizing motions with text. With the rapid development of diffusion models, a lot of works [292]–[297] introduce diffusion models into the task and achieve good generation quality. Similar to the text-to-motion task [298]–[303], the technical roadmap of music-driven dance generation can also be divided into cVAE [304], VQ-VQE [305], and diffusion-based [306], [307] methods.

IV. DATASETS AND EVALUATIONS

In this section, we summarize the commonly-used datasets for 2D, video, 3D, and 4D generation in Table VIII. Then, we present a unified and comprehensive summary of evaluation metrics in Table IX. For quantitative analysis, we evaluate metrics from two perspectives: 1) *Quality*: assessing the perceptual quality of the synthesized data, independent of input conditions (e.g., text prompts). 2) *Alignment*: measuring condition consistency, i.e., how well the generated data matches the user’s intended input. For qualitative analysis, the visual quality of generated results plays a critical role in assessing methods. Therefore, we include some human preference-based indicators as references for more effectively conducting user studies, enabling more convincing qualitative analysis results. Besides, we advocate emphasizing the practical challenges in deploying generative models, especially those related to computational efficiency. Many state-of-the-art methods require extensive GPU resources and long inference times, which hinder their accessibility and scalability in real-world applications. These factors are not always reflected in evaluation metrics but are essential when considering deployment in resource-constrained environments or interactive systems. We also encourage future benchmarks to include runtime, memory usage, and training cost to better reflect the practical feasibility of generative models.

V. FUTURE DIRECTIONS

Despite the rapid progress in 2D, video, 3D, and 4D generation, many open questions remain, especially when multiple modalities interact. These difficulties are compounded by the complex nature of integrating spatial and temporal dimensions. In 2D generation, key hurdles remain in improving realism and the diversity of generated images. For video generation, modeling long-term temporal dynamics and ensuring smooth transitions between frames are critical challenges. In 3D generation, balancing high-quality output with computational efficiency remains a key issue. Addressing these challenges is essential for advancing 4D generation, as it builds upon these established principles. Therefore, a clear roadmap is helpful not only for 4D research but also for the related 2D, video, and 3D areas that 4D models build upon.

Multimodal Generation: Generating diverse and realistic 4D content is difficult because real scenes span many modes

across space, time, and sensory channels. Findings from multimodal learning highlight three persistent barriers. First, reliable alignment across modalities is required so that the synthesised 4D asset follows the given text, image, or video prompt. Second, well-annotated multimodal datasets are scarce, especially in specialised domains, which limits the coverage of real-world dynamics that current models can learn. Third, existing architectures struggle to scale to the memory and computation cost of modelling high-resolution 3D scenes that evolve over time. Conditional diffusion, compositional latent spaces, and cross-modal contrastive pre-training have improved quality, yet achieving both diversity and fidelity under these constraints remains an open research question.

Temporal Consistency and Coherence: Ensuring smooth and realistic transitions between frames over time is a significant challenge in 4D generation. Unlike static 3D generation, 4D generation requires maintaining consistency in shape, texture, and motion across time steps. Artifacts such as flickering or unnatural deformations can easily arise, especially in long sequences. Developing methods that enforce temporal coherence without sacrificing detail or realism remains an open problem.

Efficiency and Controllability: Generating 4D assets entails large spatiotemporal tensors and lengthy optimization cycles that strain GPU memory and compute. These costs increase when score-distillation sampling is used, because its implicit formulation makes the results highly sensitive to control signals and to the diffusion prior. Compact representations and scalable architectures are needed to lower these burdens. Meanwhile, current pipelines still allow only coarse control; precise motion paths and semantic edits remain difficult. Progress will require control-oriented algorithms and clear interactive interfaces.

Fidelity and Diversity: Maintaining both fidelity and diversity in 4D generation remains difficult. For fidelity, shape, texture, and motion must stay consistent across time; otherwise, long sequences suffer from flicker, blurring of fine details, and geometric or photometric drift. Diffusion models used in image-to-4D, video-to-4D, and 3D-to-4D generation pipelines struggle to infer missing views and high-frequency detail from limited input, revealing a need for stronger temporal regularization, view-aware conditioning, and memory-efficient architectures that preserve per-frame quality while ensuring smooth evolution. For diversity, current models generalize poorly across varied objects, motion styles, and environments because dynamic 3D content is highly variable and training datasets remain narrow. Broader data coverage, adaptive conditioning, and modular training strategies will be required to achieve robust generalization, so that high-fidelity synthesis is maintained even in unseen scenarios.

Physics and Dynamics Modeling: Realistic 4D generation must reproduce collisions, elastic and plastic deformation, and fluid flow with physical accuracy, yet embedding such physics into generative models is challenging because it involves solving partial differential equations or simulating interactions in real time. This makes it hard to strike a workable balance between physical fidelity and computational speed. A

TABLE VIII: Summary of the widely-used 2D, video, 3D and 4D generation datasets. [Link] directs to dataset websites.

Dataset	Year	# Categories	# Objects	# Images	Type	Evaluation Metrics	Contribution Highlights	Limitations
2D Generation (Sec. III-A)								
SRU [308] [Link]	2011	89	-	1M	Image-Text pairs	-	Photographs with captions and 5 kinds of image content.	Noisy user captions and single reference.
MS-COCO [309] [Link]	2014	80 for objects, 91 for stuffs	1.5M	330K	Image-Text pairs	FID, CLIP-Score, SSIM	Manual annotation and 330K for English.	Class imbalance and social bias.
CC-3M [310] [Link]	2018	-	-	3M	Image-Text pairs	-	Alt-text caption annotation.	Noisy user captions and weakly aligned.
LAION-5B [187] [Link]	2022	-	-	5.85B	Image-Text pairs	FID, CLIP-Score, SSIM	5.85B CLIP-filtered multilingual image-text pairs.	Uncurated and unsafe contents.
ShareGPT4V [311] [Link]	2023	-	-	102K	Image-Text pairs	-	1.2M image-text pairs for ShareGPT4V-PT.	AI generated captions with hallucination.
Pick-a-Pic [312] [Link]	2023	-	-	500K	Image-Text pairs	FID, PickScore	Human preferences on model-generated images.	Limited preference labels.
Video Generation (Sec. III-B)								
UCF-101 [313] [Link]	2012	101	-	13K	action recognition	FVD	Manual caption annotation.	Strong background bias misleads action recognition.
ActivityNet [314] [Link]	2015	200	-	28K	action recognition	FVD	Manual caption annotation.	Background bias.
MSR-VTT [315] [Link]	2016	20	-	10K	video-language dataset	FVD, CLIP-Score	Manual caption annotation. Resolution: 240p.	Noisy and duplicate captions.
HowTo100M [316] [Link]	2019	12	154	136M	video-language dataset	FVD, CLIP-Score	ASR annotation. Resolution: 240p.	Noisy and misaligned narration.
WebVideo-10M [317] [Link]	2021	-	-	10M	video-language dataset	FVD, CLIP-Score	Alt-text caption annotation. Resolution: 360p.	Low-resolution and watermarked videos.
HD-VILA-100M [318] [Link]	2022	15	-	103M	video-language dataset	FVD, CLIP-Score	ASR annotation. Resolution: 720p.	Noisy and misaligned ASR subtitles.
InternVid [319] [Link]	2023	16	-	7M	video-language dataset	FVD, CLIP-Score	Automatic caption annotation.	Limited diversity.
Panda-70M [188] [Link]	2024	-	-	70M	video-language dataset	FVD, CLIP-Score	Automatic caption annotation. Resolution: 720p.	Limited diversity.
Koala-36M [320] [Link]	2024	-	-	36M	video-language dataset	FVD, CLIP-Score	Automatic and manual caption annotation. Resolution: 720p.	Inherent biases from its source corpus.
3D Generation (Sec. III-C)								
DeepFashion [321] [Link]	2016	50	300K	800K	single-view images	-	Human clothing dataset.	No 3D data.
SHHQ [322] [Link]	2022	-	-	230K	single-view images	-	Human body dataset.	(1) No 3D data. (2) Inherent biases from FFHQ.
CO3D [323] [Link]	2021	50	19K	1.5M	multi-view images	PSNR/LPIPS; IoU, etc.	Annotated with camera poses and point clouds.	Lack high-fidelity geometry/textures.
RTMV [324] [Link]	2021	-	2K	300K	multi-view images	PSNR/LPIPS; MAE, etc.	Large-scale synthetic dataset with high-resolution images.	Synthetic data cannot fully capture real-world complexity.
MVImgNet [325] [Link]	2023	238	219K	6.5M	multi-view images	PSNR/LPIPS	Annotated with camera poses and point clouds.	Lack high-fidelity geometry/textures.
uCO3D [326] [Link]	2025	1K	170K	170K	multi-view images	PSNR/LPIPS; IoU	Annotated with camera poses, point clouds, depth maps, caption, and 3DGS.	Lack high-fidelity geometry/textures.
ShapeNet [327] [Link]	2015	3K	51K	-	3D data	-	Large-scale synthetic CAD dataset.	Synthetic data cannot fully capture real-world complexity.
3D-Future [328] [Link]	2020	34	10K	-	3D data	CLIP Similarity	High-quality 3D models with textures and attributes.	(1) Lack object diversity. (2) Synthetic data cannot fully capture real-world complexity.
GSO [183] [Link]	2022	17	1K	-	3D data	F-Score/ CLIP Similarity	High-fidelity geometry/textures	Limited to household objects.
Objaverse [185] [Link]	2022	21K	818K	-	3D data	F-Score/ CLIP Similarity	Large-scale dataset.	Variable data quality.
Objaverse-XL [186] [Link]	2023	-	10.2M	-	3D data	F-Score/ CLIP Similarity	Large scale and high diversity.	Variable data quality.
Cap3D [329] [Link]	2023	-	785K	-	3D-text pairs	FID/CLIP etc.	Rich text annotations	Inherent bias in captions from pretrained models.
Animal2400 [132] [Link]	2023	-	2400	-	prompts	CLIP R-Precision	10 animals, 8 activities, 6 themes, 5 hats.	Limited object diversity.
DF27/DF415 [142] [Link]	2024	-	27/415	-	prompts	CLIP R-Precision	-	Limited to synthetic or artist-created objects.
4D Generation (Sec. III-D)								
Consistent4D [245] [Link]	2023	-	-	12(in-the-wild), 14(synthetic)	multi-view videos	LPIPS/CLIP	Video-to-4D Generation.	-
Diffusion4D [231] [Link]	2024	-	365K	365K	dynamic 3D data	-	Collected from Objaverse-1.0 (42K) and Objaverse-xl (323K).	-
MV-Video [229] [Link]	2024	-	53K	53K	dynamic 3D data	-	Annotated by MiniGPT4-Video [330]	Inaccurate captions.
CamVid-30K [233] [Link]	2024	-	-	30K	4D data	-	Real-world 4D scene dataset	-
4D-DRESS [331] [Link]	2024	-	-	64(outfits)	human clothing 4D data	CD	Total number of 3D human frames: 78k.	Manual-heavy pipeline limits scalability.

"ASR" stands for automatic speech recognition, "pcl" for point clouds, and "CD" for Chamfer Distance.

TABLE IX: Summary of common evaluation metrics.

Notation	Description
Quality (quantitative analysis)	
Image-level	PSNR $\uparrow$ <i>Image Fidelity/Similarity. Peak Signal-to-Noise Ratio: the ratio between the peak signal and the Mean Squared Error (MSE).</i>
	SSIM $\uparrow$ <i>Image Fidelity/Similarity. Structural Similarity Index Measure [332]: evaluating brightness, contrast, and structural features between generated and original images.</i>
	LPIPS $\downarrow$ <i>Image Fidelity/Similarity. Learned Perceptual Image Patch Similarity [333]: metric computed with a model trained on labeled human-judged perceptual similarity.</i>
	FID $\downarrow$ <i>Image Fidelity/Similarity. Fréchet Inception Distance [84]: comparison of generated and GT image distributions.</i>
	IS $\uparrow$ <i>Image Fidelity/Similarity. Inception Score [334]: metric only evaluated image distributions and calculated by pretrained Inception v3 model.</i>
Video-level	FVD $\downarrow$ <i>Video Fidelity. Fréchet Video Distance [335]: the Inflated-3D Convnets (I3D) pretrained model to compute their means and covariance matrices for scores.</i>
	KVD $\downarrow$ <i>Video Fidelity. Kernel Video Distance [335]: an alternative to FVD proposed in the same work, using a polynomial kernel.</i>
	Video IS $\uparrow$ <i>Video Fidelity. Video Inception Score [336]: the inception score of videos with the features extracted from C3D [337].</i>
	FVMD $\downarrow$ <i>Video Fidelity. Fréchet Video Motion Distance [338]: metric focused on temporal consistency, measuring the similarity between motion features of generated and reference videos using Fréchet Distance.</i>
Alignment (quantitative analysis)	
Semantics	CLIP Similarity [339] $\uparrow$ <i>Text-Image Alignment. Same as CLIP-Score, defined as a text-to-image similarity metric, measuring the (cosine) similarity of the embedded image and text prompt.</i>
	CLIP R-Precision [340] $\uparrow$ <i>Text-Image Alignment. The CLIP model's accuracy at classifying the correct text input of a rendered image from amongst a set of distractor query prompts.</i>
	X-CLIP [341] $\uparrow$ <i>Text-Image Alignment. Measured by a video-based CLIP model finetuned on text-video data.</i>
	CLIP-T $\uparrow$ <i>CLIP Textual Alignment. The average cosine similarity between the generated frames and text prompt with CLIP ViT-B/32 [342] image and text models.</i>
	CLIP-I $\uparrow$ <i>CLIP Image Alignment. The average cosine similarity between the generated frames and subject images with CLIP ViT-B/32 image model.</i>
	DINO-I $\uparrow$ <i>DINO [86] Image Alignment. The average visual similarity between generated frames and reference images with DINO ViT-S/16 model.</i>
User study (qualitative analysis)	
AQ $\uparrow$	<i>Human Preference. Appearance Quality: percentage unit (%).</i>
GQ $\uparrow$	<i>Human Preference. Geometry Quality: percentage unit (%).</i>
DQ $\uparrow$	<i>Human Preference. Dynamics Quality: percentage unit (%).</i>
TA $\uparrow$	<i>Human Preference. Text Alignment: percentage unit (%).</i>

promising way forward is to pair generative networks with differentiable simulators or neural ordinary differential equation layers, letting gradient-based training enforce Newtonian laws. Building such coupled systems forms the basis of a world model that encapsulates real-world characteristics, providing a natural benchmark for evaluating physically and dynamically consistent 4D generation. Future research can build on these benchmarks to better simulate realistic interactions while maintaining computational efficiency.

VI. CONCLUSIONS

In this survey, we have reviewed recent advances in multi-modal generative models for simulating the real world, focusing on the intertwined dimensions of appearance, dynamics, and geometry. We categorized existing approaches across 2D, video, 3D, and 4D generation, and discussed their representative methodologies, cross-domain relations, and technical dif-

ferences, supported by comparative visual examples. Furthermore, we summarized commonly used datasets and evaluation metrics to provide a practical reference for benchmarking.

Despite the rapid progress, fundamental challenges persist, such as the scalability of generative pipelines, temporal consistency in long-range sequences, and adaptability to real-world dynamics. We identified several open research directions, including the need for unified representations across modalities, efficient training with sparse supervision, and the integration of physics-based constraints for enhanced realism.

We hope this survey provides not only a comprehensive overview for newcomers but also a foundation for future research towards more coherent, controllable, and physically grounded multimodal generation systems.

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